Vector Search Primer (public-domain fixture, written for NEXUS Local)

Approximate nearest neighbor search finds vectors close to a query vector without scanning the entire index. HNSW builds a multi-layer proximity graph: upper layers provide long-range links for fast navigation while the bottom layer contains every vector.

Recall measures the fraction of true nearest neighbors returned. Higher ef_search values improve recall at the cost of latency.

Page 2: Hybrid Retrieval

Hybrid retrieval combines dense vector similarity with lexical BM25 scoring. Reciprocal Rank Fusion merges the two rankings using the formula score = sum(1 / (k + rank)) with k commonly set to 60. Fusion is robust because it depends only on ranks, not raw scores.